PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AGENTIC MULTI-TENANT MDR FRAMEWORK FOR AI-ASSISTED SECURITY OPERATIONS

AN INTERNSHIP REPORT

Submitted by

HIDA FATHIMA P H - 20221CAI0002

Under the guidance of,

Dr. VIJAYAKUMAR ADAICKALAM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence & Machine Learning)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"Agentic Multi-Tenant MDR Framework For AI-Assisted Security Operations"** is a bonafide work of **HIDA FATHIMA P H (20221CAI0002),** who has successfully carried out the internship work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION), ARTIFICIAL INTELLIGENCE & MACHINE LEARNING during 2025-2026.

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1.	MOHAMMADI AKHEELA KHANUM	21.4.26
2	Dr. Rajesh C.B	

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING), at Presidency University, Bengaluru, named **Hida Fathima P H**, hereby declare that the internship work titled **"Agentic Multi-Tenant MDR Framework For AI-Assisted Security Operations"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Spl. Division) (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Hida Fathima P H USN: 20221CAI0002 Hida

PLACE: BENGALURU

DATE: 20 April 2026

INTERNSHIP COMPLETION CERTIFICATE

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Date: April 10, 2026

Subject: Internship Experience Letter

TO WHOMSOEVER IT MAY CONCERN

This letter is to certify that **Ms. Hida Fathima PH** was working as Intern - AI and Cyber Security with Tecplix Technologies Pvt. Ltd. Her internship tenure was from January 05, 2026 to April 03, 2026.

We wish her all the best for her future endeavors.

For Tecplix Technologies Pvt Ltd

TECPLIX TECHNOLOGIES PVT. LTD.

Sahana P S
Sr. Vice President-HR

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I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering (Spl. Division), and also the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

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Hida Fathima P H

ABSTRACT

The current problem that Security Operations Centers (SOCs) are grappling with is a huge data issue. Every day, modern enterprise networks are producing thousands of security alerts, and the human analysts cannot possibly go through them all manually. During the course of this internship, as I learned about SOC workflows, it became apparent that there is a great deal of time wasted by analysts who have to switch between unrelated security monitoring tools. This swivel chair analysis has been repeated to cause severe alert fatigue and significantly slows down the speed at which a team can halt a real cyber-attack. Although numerous companies are trying to apply Large Language Models (LLMs) to accelerate the process of alert triage, the upload of sensitive company network logs to AI models in the public clouds poses unacceptable risks to data privacy. Moreover, common LLMs also hallucinate fabricated facts, which is highly perilous during incident response.

The proposed project proposes the Agentic Multi-Tenant Managed Detection and Response (MDR) Framework to address these practical problems. The system was configured as a dockerized microservice environment that does not use conventional chatbot interfaces by deploying a 7-Agent AI Swarm which runs locally. Since, the LLM inferencing occurs on the host machine itself, there is no risk of corporate information being leaked into external servers. The platform reports live attacks to an external decoy honeypot sensor and feeds which are then fed to a localized SQLite data lake via zero-trust webhooks. The framework will support various security personnel members with a severe 14-persona Role-Based Access Control system, meaning that users can only interact with the particular AI agents and data as authorized by their job role. In the course of an investigation, the swarm employs Retrieval-Augmented Generation and local threat heuristics to process logs, identify technical artifacts, correlate attacks with the MITRE ATT&CK framework and generate formal SOAR incident reports automatically.

Experiments showed the division of the AI into specialized tasks and rigid restrictions effectively prevented the creation of fake data by the models. Finally, this framework offers a safe, operational remedy that vastly saves the Mean Time to Respond (MTTR) to threats, and yet preserves human analyst in the ultimate decision-making of security.